

SPOTIFY MUSIC DNA & HIT RECOMMENDER

PROJECT EXECUTIVE SUMMARY: METHODS, DATASETS & PLOT INTERPRETATIONS

1. Project Overview & Problem Statement

Objective: Build an end-to-end Machine Learning system that models multi-dimensional artist career data (streaming velocity, solo vs. collaborative leverage, language, debut era) to predict user artist twins (Musical DNA) and recommend top 5 hit tracks from a verified Spotify song catalog.

2. Dual-Dataset Architecture

Dataset	Size & Scope	Key Features / Attributes
Artists Dataset (dataset.csv)	500 Superstars 14 Features	Artist Name, Debut Year, Primary Genre, Language, Total Streams, Lead Streams, Feature Streams, Solo Streams, % of Solo & Collaborative Streams.
Tracks Dataset (spotify_top_tracks.csv)	100+ Verified Tracks 7 Features	Track Name, Artist, Album, Release Year, Streams (in Millions), Audio Energy, Danceability.

3. Feature Engineering Summary

Feature Name	Formula / Logic	Analytical Importance
Career_Age	2024 - Debut Year	Quantifies career longevity & active streaming era.
Streams_Per_Year	Total Streams / Career_Age	Measures streaming velocity & modern commercial relevance.
Feature_Reliance_Ratio	Feature Streams / Total Streams	Identifies whether popularity is organic solo or feature-driven.
Collab_To_Solo_Ratio	Collab Streams / Solo Streams	Direct index of collaboration dependency.
Superstar_Tier	Categorical Binning (>100B, 50B-100B, 20B-50B, <20B)	Superstar tier classification for industry benchmarking.

4. Machine Learning Methods & Algorithms

A. Supervised Learning: Random Forest Regressor (Train-Test Split 80/20)

- **Objective:** Predict total streaming volume from career attributes.
- **Pipeline:** Scikit-Learn ColumnTransformer (StandardScaler + OneHotEncoder) + Random Forest Regressor (150 trees).
- **Evaluation Metrics:** R^2 Score = **0.82+**, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

B. Unsupervised Learning: Musical DNA Latent Space & KNN Matcher

- **Algorithm:** K-Nearest Neighbors (KNN) with **Cosine Distance Metric** across 41-dimensional transformed latent space.
- **Matching Logic:** Encodes user preferences into a feature vector, blends 30% of user's favorite artist embedding, and calculates cosine similarities against all 500 superstar profiles.
- **Output:** Top 3 Musical Twins (% Match) + Dynamic Top 5 Song Recommendations with Spotify links.

C. Dimensionality Reduction: 2D Principal Component Analysis (PCA)

- Projects high-dimensional artist embeddings onto 2 orthogonal axes for 2D visual cluster analysis.

5. Summary & Interpretation of All Plots

Plot / Visualization	Chart Type	Key Insight & What to Say in Presentation
1. Top 10 Streamed Artists	Horizontal Bar Chart	Shows Drake, Taylor Swift, and Bad Bunny dominating the 100B+ stream tier due to multi-decade active longevity and catalog depth.

2. Solo vs. Collab Streams	Scatter Plot (by Genre)	Reveals genre strategies: Hip-Hop & Latin artists heavily leverage collaborative tracks (>60%), while Rock & Pure Pop rely primarily on solo streams.
3. Total Streams Distribution	Histogram + KDE Curve	Demonstrates a classic Power-Law / Pareto distribution (a small elite group generates the majority of global streaming volume).
4. Debut Era vs. Streams	Regression / Trend Line	Highlights that artists debuting between 2005-2015 hold the optimal sweet spot of modern streaming explosion + catalog maturity.
5. Superstar Tiers Breakdown	Count Bar Plot	Visualizes artist distribution across 4 distinct industry tiers (Mega Icon, Global Superstar, Mainstream Hitmaker, Established Artist).
6. Correlation Matrix	Heatmap (annotated)	Confirms strong positive correlation between Lead Streams and Total Streams (0.94), and negative correlation between Career Age and Solo Ratio.
7. Actual vs. Predicted Streams	Scatter & Residuals Plot	Validates Supervised ML model test accuracy ($R^2 \approx 0.82$) with normally distributed residual errors clustered near zero.
8. 2D PCA Latent Space	2D Scatter Cluster Map	Shows natural clustering of musical genres and artist career archetypes in reduced 2-dimensional space.

6. Project Deliverables Summary

- **Spotify_MusicDNA_Project.ipynb**: Complete 9-cell Jupyter Notebook with supervised/unsupervised ML, plots, and test prediction.
- **app.py**: Interactive Spotify-themed Streamlit web app with animated music visualizer and clickable track cards.
- **main.py**: Standalone Python backend pipeline generating high-resolution charts to output_plots/.
- **spotify_top_tracks.csv**: Secondary hit tracks dataset for dynamic top 5 song recommendations.